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***Abstract*—This paper presents a cyber-physical surveillance system designed to monitor both digital and physical security layers. The system is composed of two independent modules. The first addresses anomaly detection in Windows endpoint logs by comparing a classical machine learning pipeline, based on Isolation Forest and a GRU autoencoder, against Dual Sentinel, a fully LLM-native approach that uses a small language model for first-pass triage and a larger model as a judge to validate findings and produce natural-language explanations mapped to MITRE ATT&CK techniques. The second module addresses physical access control through a face recognition pipeline built on deep embeddings and cosine similarity matching, following an open-set verification paradigm. Both modules are evaluated on public datasets. The comparison between classical and LLM-native detection explores the trade-off between raw performance and interpretability, a gap that remains underexplored in literature. Results and limitations of each approach are discussed.**

Keywords— anomaly detection, Windows endpoint logs, Sysmon, large language models, small language models, LLM-as-a-Judge, MITRE ATT&CK, Isolation Forest, GRU autoencoder, face recognition, open-set verification, cyber-physical security.

I. Introduction

The convergence of digital and physical security has emerged as a critical challenge in the design of modern protection systems. Traditional security architectures typically treat network intrusion detection and physical access control as independent domains. However, contemporary threats increasingly span both layers, where adversaries may exploit software vulnerabilities to gain initial access and subsequently manipulate physical systems such as surveillance infrastructure or access control mechanisms. Addressing this paradigm requires integrated cyber-physical surveillance systems capable of jointly monitoring digital host behavior and physical presence within protected environments.

This work presents the design, implementation, and evaluation of a unified cyber-physical surveillance system composed of two complementary modules. The first focuses on the digital domain by performing anomaly detection over Windows endpoint logs, aiming to identify suspicious or malicious activity within monitored systems. The second operates in the physical domain through a face recognition pipeline that identifies individuals captured by cameras and verifies their authorization status. While these modules operate independently, they share a common objective of providing a holistic view of threats across both cyber and physical dimensions.

The digital component of this system builds upon recent advances in log-based anomaly detection for Windows environments. Windows telemetry sources such as Event Tracing for Windows (ETW) and Sysmon provide high-fidelity behavioral data, including process creation, network connections, file operations, and registry modifications [9]. These logs form the foundation for many modern detection systems, as demonstrated in recent datasets and tools such as LMD-2023 and AAU_MalData [1], [7]. However, the volume and complexity of such data pose a significant challenge, as a single host may generate millions of events per day, most of which correspond to benign activity.

To address this challenge, prior work has extensively explored machine learning (ML) and deep learning (DL) techniques for anomaly detection. Classical approaches typically rely on structured feature extraction followed by models such as Isolation Forest, Support Vector Machines, or tree-based ensembles. Notably, studies have shown that well-engineered features combined with ensemble methods (e.g., Extra Trees) can achieve near state-of-the-art performance, with reported F1 scores above 0.99 in lateral movement detection tasks [1], [2]. In parallel, sequence-based models such as LSTM and CNN–LSTM hybrids have been proposed to capture temporal dependencies in event streams, achieving similarly high performance in ransomware detection scenarios [4], [5].

Despite these advances, several limitations remain. First, anomaly detection models are inherently statistical and lack semantic interpretability, producing scores that do not directly explain the underlying behavior or map to known attack techniques. Second, most studies rely on controlled datasets, with limited evaluation of concept drift, where normal system behavior evolves over time and degrades model performance [5]. Third, the cost of false positives and the operational burden on analysts are rarely quantified, despite being critical factors in real-world

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deployments [13]. Additionally, data scarcity and limited availability of public datasets continue to hinder reproducibility and generalization across environments [18].

Recent research has begun to explore the use of Large Language Models (LLMs) for cybersecurity tasks, leveraging their ability to reason over structured textual data and generate human-readable explanations. Unlike traditional models, LLMs can analyze sequences of events, relate them to known attack patterns such as those defined in the MITRE ATT&CK framework [14], and provide contextualized interpretations of suspicious behavior. However, LLMs are also prone to hallucinations, generating plausible but unsupported claims, which introduces risks in security-critical applications.

To mitigate these limitations, this work adopts a Dual LLM architecture in which one model performs initial anomaly assessment while a second model acts as a judge, validating the reasoning against observed evidence. This approach aligns with recent trends in LLM-as-a-judge frameworks, which improve reliability through cross-validation of model outputs [17]. The system integrates this architecture with classical anomaly detection models, forming a hybrid pipeline that combines statistical detection with semantic reasoning.

The proposed system is evaluated using a diverse set of public datasets, including lateral movement (LMD-2023), ransomware detection datasets, and ETW-based traces, complemented by baseline logs collected under normal operating conditions. Evaluation considers both traditional metrics, such as F1-score and false-positive rate, and LLM-specific dimensions, including explanation quality, evidence grounding, and hallucination rate.

II. Related Work

The detection of advanced cyber threats, lateral movement, and ransomware operations has undergone a profound and multifaceted methodological evolution over the past half-decade. As sophisticated adversaries increasingly leverage "living-off-the-land" (LotL) techniques, fileless execution, and dynamically encrypted payloads, traditional signature-based intrusion detection systems and static heuristic engines have been rendered largely obsolete. In response, the academic, industrial, and cyber-defense research communities have pursued increasingly complex behavioral modeling paradigms, transitioning from localized statistical analyses to highly contextualized, semantically aware reasoning engines.

This section provides an exhaustive, chronologically, and thematically structured examination of the literature that underpins modern cybersecurity anomaly detection. (on work)

A. Windows Telemetry and Log Sources

Sysmon and ETW are the dominant sources of behavioral telemetry on Windows hosts, recording process, network, file, and registry activity at fine granularity. Kellect, introduced by Chen et al., provides a kernel-based ETW collector with minimal data loss [1]. The LMD-2023 dataset, built from real Sysmon logs, serves as the primary benchmark in this work [2]. The volume of events per host per day, often in the millions, makes isolating malicious activity a significant signal-to-noise problem.

B. Classical Machine Learning for Log Anomaly Detection

Feature-based ML methods extract numeric statistics from fixed time windows and apply classifiers or anomaly detectors. Smiliotopoulos and Kambourakis demonstrated that well-engineered Sysmon features with Extra Trees reach AUC above 0.99 on lateral movement detection [2], [3], with follow-up work showing the significant impact of class imbalance handling on reported results [3], [4]. Achmad et al. applied LOF with PCA to malware classification, reaching F1 above 0.98 [5], and Gwak et al. explored explainable ETW-based classification, highlighting the interpretability gap between model outputs and analyst understanding [6]. Isolation Forest, proposed by Liu et al., is widely used in this setting as a one-class detector requiring no labeled attack data [7]. The shared limitation across all these approaches is that outputs are scores without semantic content, giving analysts no direct insight into what behavior was detected or which technique it corresponds to.

C. Sequence-Based and Autoencoder

Temporal models learn patterns across ordered event sequences. Ispahany et al. achieved F1 above 0.99 on incremental ransomware detection in Sysmon streams [8], and their SILRAD system added adaptive drift detection via ADWIN to maintain performance as normal behavior evolves [9]. Guo et al. showed that correlating performance counters over a six-minute window achieves 100% TPR at low false positive rates for ransomware [10]. Autoencoder architectures trained on normal-only data detect anomalies through reconstruction error and have been applied in critical infrastructure [11], with GRU variants proving effective where temporal event structure is informative [12]. Darban et al. survey the full design space of deep learning methods for time series anomaly detection [13]. The GRU autoencoder in the classical system evaluated here follows this one-class training paradigm.

D. Provenance Graph-Based Detection

The Provenance graph methods represent host activity as graphs of causal relationships between system entities. ProGrapher applies graph neural networks to detect anomalous subgraphs [14], and ORTHRUS extends this with temporal graph learning for higher detection fidelity [15]. Despite strong results, Abrar et al. found significant reproducibility concerns across this class of systems, with most results depending on proprietary datasets or configurations [16]. This gap, combined with the operational cost of maintaining full provenance capture, motivates the simpler log-based approach used here.

E. Large Language Models for Cybersecurity

LLMs can process log events as text and produce natural-language descriptions of observed behavior, enabling direct mapping to MITRE ATT&CK techniques [17] in a way that statistical models cannot. Phi-3 from Microsoft is a compact SLM designed for efficient local inference on structured reasoning tasks [18], and Meta's Llama 3.2 covers a range of sizes suited to edge deployment [19]. Running both models locally via Ollama [20] avoids

transmitting sensitive log content to external APIs, which is a practical requirement in security deployments.

F. LLM-as-a-Judge

The LLM-as-a-Judge framework uses a second model to validate the output of a first, reducing dependence on human annotation at inference time. Gu et al. survey evaluation criteria and known failure modes including positional and verbosity bias [21]. G-Eval showed that explicit scoring rubrics bring GPT-4 judgments close to human agreement [22], and Zheng et al. benchmarked judge reliability across model pairs using MT-Bench and Chatbot Arena [23]. In Dual Sentinel, Phi-3 produces an initial verdict per log window and Llama 3.2 evaluates that verdict against the observed evidence before the final result is issued.

G. Hallucination and Prompt Injection in LLM Security Systems

Hallucination, where a model generates confident but unsupported claims, is a specific risk in security contexts. Preuveneers et al. identified conditions under which LLMs fabricate ATT&CK technique identifiers or misattribute behavior [24], and Alansari and Luqman provide a broader taxonomy of hallucination types and mitigations [25]. Prompt injection, where adversarial content in the input manipulates model output, is classified by OWASP as the primary vulnerability for LLM-integrated applications [26]. Liu et al. demonstrated that short injected payloads can alter model verdicts in deployed systems [27]. In Dual Sentinel, the judge's evidence grounding check is expected to catch both fabricated claims from the analyst and injected instructions that were not present in its own input.

H. Deep Face Recognition

Face recognition systems follow a shared pipeline: detect and align the face, extract an embedding, and match against enrolled identities. ArcFace established the angular margin loss function that is now the standard training objective for discriminative embeddings [30]. MTCNN by Zhang et al. introduced multitask cascaded detection and landmark localization for consistent geometric alignment [28], extended by InsightFace's RetinaFace detector which improves robustness and CPU efficiency [29], [31]. ResNet-50 with ArcFace is the backbone configuration used in this work [33]. Fuad et al. survey the evolution of deep face recognition architectures and loss functions [32], and the augmentation strategy used during training follows established practices for robustness to lighting, occlusion, and image quality variation [34].

I. Open-Set Recognition and Anti-Spoofing

In the closed-set paradigm, identity decisions are made by a softmax classifier over a fixed set of known individuals, which requires retraining on enrollment changes and cannot reliably reject unknown inputs. The open-set paradigm instead compares an embedding against enrolled gallery centroids using a similarity threshold, supporting dynamic enrollment and natural rejection of unknowns. Scheirer et al. formalized this problem and showed that closed-set classifiers systematically overestimate confidence on unknown inputs [36], motivating the embedding-based approach used here. Presentation attacks, where an attacker uses a photo or video to spoof the system, remain an open problem. Yu et al. survey anti-spoofing methods across texture analysis, depth sensing, and challenge-response approaches, noting that no method generalizes robustly across unseen attack types [35]. The absence of liveness detection in the current system is acknowledged in Section V.

III. System Architecture

A. Classical Anomaly Detection Pipeline

B. Dual LLM Reasoning Module (DualSentinel)

DualSentinel delegates anomaly assessment to two locally hosted language models operating in a sequential analyst–judge configuration (Fig. X). Parsed and windowed Sysmon logs are passed directly to the LLM pipeline, where a rule-based ATT&CK tagger first annotates each window and a structured Evidence Pack is assembled for model consumption.

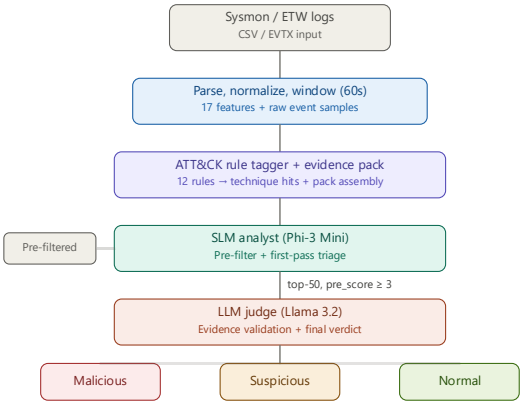

Fig. 1. Example of a figure caption. (*figure caption*)

Both models run entirely on-premise via Ollama [20], avoiding the transmission of sensitive telemetry to external services. The SLM Analyst uses Microsoft Phi-3 Mini, a compact model with approximately 4 billion parameters [18], for rapid triage. The LLM Judge uses Meta Llama 3.2, with approximately 8 billion parameters [19], for authoritative verdict generation. Both models are configured with a temperature of 0.1 and JSON-only output formatting enforced at the API level.

1) Evidence Pack

Each window is represented as a structured Evidence Pack containing only observable facts: aggregate statistics (event counts, unique processes, entropy values), pre-computed ATT&CK rule tagger hits with associated confidence scores, and a representative sample of individual events. The rule tagger evaluates twelve hand-coded conditions that map features such as PowerShell execution, lateral movement port activity, and process injection indicators to specific ATT&CK technique identifiers (e.g., T1059.001, T1021.002, T1055). The sample size adapts to window density, including up to 15, 30, or 50 event lines for small, medium, and large windows, respectively. Command-line strings are truncated to 120 characters and code fences are sanitized to mitigate prompt injection. The pack contains no interpretive content, ensuring that each model reasons exclusively from raw observations.

2) SLM Analyst

The Analyst identifies risk indicators visible in the data, suggests potential ATT&CK techniques, assigns a preliminary risk score on a 0–10 integer scale, and determines whether the window warrants deeper analysis by the Judge. Its system prompt explicitly prohibits the invention of process names, IP addresses, or file paths not present in the Evidence Pack. A deterministic pre-filter operates before the Ollama call: windows exhibiting zero threat signals — defined as having no suspicious processes, no shell activity, no tool indicators (e.g., Mimikatz, PsExec), no ATT&CK rule hits, and fewer than five network connections — bypass the model entirely and are assigned a pre-score of zero. On the LMD-2023 normal baseline comprising 1,424 windows, this mechanism eliminated 500 windows (35.1%), reducing the total number of inference calls to 924.

3) LLM Judge

The Judge receives both the original Evidence Pack and the Analyst's pre-diagnosis, which is explicitly framed in the prompt as a hypothesis to validate rather than an established fact. For every ATT&CK technique it reports, the Judge must cite a specific event or statistic from the Evidence Pack. Rule tagger hits that lack corroborating evidence in the pack are downgraded to low confidence. The scoring rubric, embedded directly in the system prompt, follows a defined scale: 0–2 corresponds to normal or benign activity, 3–4 to mildly suspicious behavior, 5–6 to moderately suspicious activity with lateral movement indicators, 7–8 to highly suspicious findings with confirmed indicators such as Mimikatz or PsExec, and 9–10 to critical detections with multiple corroborated ATT&CK technique mappings. The Judge also reports any Analyst claims it could not substantiate in a dedicated unsupported_claims field. A cap of 50 windows is applied to the Judge stage, selected by descending Analyst pre-score. Windows where the Analyst assigned a pre-score below 3 and flagged no need for deep analysis are skipped entirely.

4) Hallucination Mitigation

Applying language models to security-critical analysis introduces the risk of hallucination, where a model generates plausible but unsupported claims [24], [25]. DualSentinel addresses this risk through five complementary mechanisms. First, the Evidence Pack is constructed to contain only verifiable observations, providing no interpretive content that a model could distort. Second, both system prompts include explicit prohibitions against fabricating identifiers not present in the input. Third, the JSON-only output format is enforced at the Ollama API level, and a three-step fallback parser (direct JSON parse, markdown fence stripping, and regex brace matching) ensures robust structured output extraction. Fourth, the Analyst's output is labeled as a hypothesis in the Judge's prompt, preventing the uncritical propagation of first-pass errors. Fifth, the unsupported_claims field creates a traceable disagreement log between the two models, supporting downstream audit and reliability assessment.

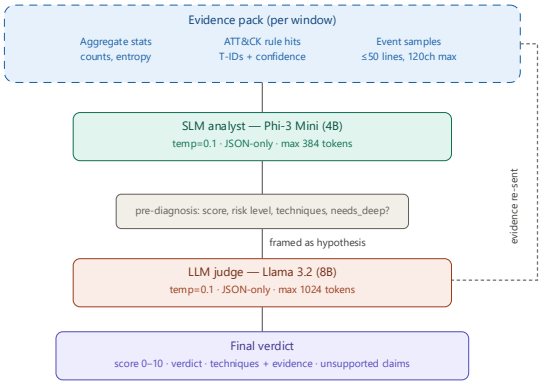

Fig. 2. Example of a figure caption. (*figure caption*)

C. Dual LLM Reasoning Module

(write to review)

D. Face Recognition Module

1) Closed-Set Training Pipeline:

The closed-set approach fine-tunes a deep model on authorized individuals and classifies identities via softmax (Fig. Z).

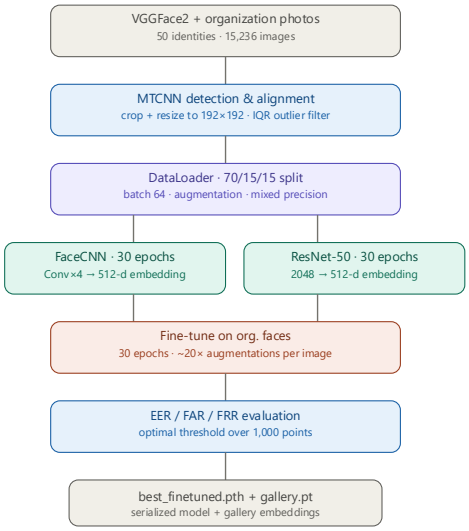

Fig. 3. Example of a figure caption. (*figure caption*)

The training set is drawn from VGGFace2 [28], with 15,236 images across 50 identities split 70/15/15. MTCNN [27] performs face detection and alignment, cropping each image to 192×192 pixels, followed by IQR-based outlier removal.

Two architectures are compared. FaceCNN is a four-block convolutional network (Conv3 × 3 → BatchNorm → ReLU → MaxPool per block) producing 512-dimensional embeddings. ResNet50Face uses a ResNet-50 backbone [31] trained from scratch, projecting its 2,048-dimensional output to 512 dimensions. Both use a softmax head during training and produce L2-normalized embeddings at inference.

Training runs for 30 epochs with Adam (lr=10^{-3}, weight decay=10^{-4}), cosine annealing, and mixed-precision. Augmentation includes random flips, color jitter,

Gaussian blur, and rotation. Both models are then fine-tuned for 30 additional epochs on organization-specific photos with a stronger augmentation regime (~20 variants per image). The optimal operating threshold is determined through FAR/FRR analysis over 1,000 points, and the best model and gallery embeddings are exported [30].

2) Open-Set Real-Time Pipeline

The open-set approach uses a pre-trained ArcFace model [29, 30] and performs verification via embedding similarity against enrolled centroids (Fig. W).

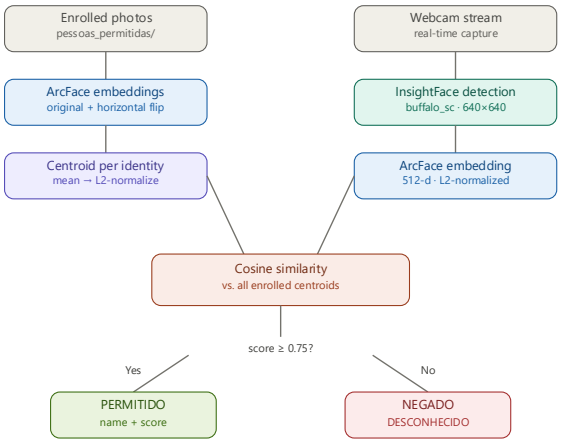

Fig. 4. Example of a figure caption. (*figure caption*)

The system uses InsightFace's buffalo_sc model served via ONNXRuntime on CPU. Detection and 512-dimensional embedding extraction occur in a single inference call. Enrollment is directory-based: reference images are loaded at startup, embeddings are extracted from each image and its horizontal flip, and a single L2-normalized centroid is computed per identity.

At inference, each webcam frame is processed in real time. The extracted embedding is compared against all centroids via cosine similarity. If the best score meets or exceeds the threshold (default 0.75, adjustable at runtime), access is granted; otherwise the individual is classified as unknown and denied. Decisions are displayed through color-coded bounding boxes overlaid on the video feed.

3) Paradigm Comparison

The closed-set approach achieves strong discriminative performance on fixed identity sets through end-to-end supervised optimization but requires retraining upon enrollment changes and lacks natural rejection of unknowns [32]. The open-set approach enables dynamic enrollment without retraining, supports unknown rejection via thresholding, and runs on CPU hardware alone. Both produce 512-dimensional embeddings, enabling direct comparison.

4) Limitations

The current system does not incorporate liveness detection or anti-spoofing mechanisms, leaving it vulnerable to presentation attacks [35]. This is discussed further in Section V.

IV. Experimental Setup and Results

After

V. Discussion and Limitations

VI. Conclusion

References

[1] S. Chen et al., "Kellect: Kernel-Based Lossless Event Log Collector," arXiv:2207.11530, 2022.

[2] C. Smiliotopoulos and G. Kambourakis, "Sysmon event logs for machine learning-based malware detection," Int. J. Inf. Secur., vol. 22, no. 7, pp. 1–16, 2023.

[3] C. Smiliotopoulos and G. Kambourakis, "Machine Learning for Lateral Movement Detection: Imbalanced vs Resampled Data," Int. J. Inf. Secur., 2026 (preprint).

[4] C. Smiliotopoulos et al., "Revisiting the detection of lateral movement through Sysmon," Appl. Sci., vol. 12, no. 15, p. 7746, 2022.

[5] R. Achmad et al., "Sysmon event logs for machine learning-based malware detection," Cyber Secur. Appl., vol. 5, p. 100110, 2025.

[6] J. Gwak et al., "Explainable Malware Classification based on ETW Logs," in Proc. CiSE-CxAI, 2023.

[7] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation Forest," in Proc. IEEE ICDM, 2008, pp. 413–422.

[8] J. Ispahany et al., "Online Batch Incremental Ransomware Detection in Sysmon Data Streams," IEEE Access, vol. 11, pp. 14111–14128, 2023.

[9] J. Ispahany et al., "SILRAD: Streaming Machine Learning for Ransomware Detection," in Proc. Annual Res. Secur. Conf., 2025.

[10] Y. Guo et al., "Ransomware Detection through Correlated Performance Counters," in Proc. ACM FSE, 2025.

[11] A. Demetriou et al., "Exploiting Autoencoder-Based Anomaly Detection to Enhance Cybersecurity in Power Grids," Future Internet, vol. 16, no. 6, p. 184, 2024.

[12] G. A. Wilkins, "Real-Time Anomaly Detection in OT Networks Using GRU-Based Autoencoders," M.S. thesis, Univ. of Arkansas, 2025.

[13] Z. Zamanzadeh Darban et al., "Deep Learning for Time Series Anomaly Detection: A Survey," ACM Comput. Surv., vol. 57, no. 1, 2024.

[14] Y. Yang et al., "ProGrapher: Graph-based Provenance Anomaly Detection," in Proc. USENIX Security, 2023.

[15] B. Jiang et al., "ORTHRUS: High-Fidelity Host Anomaly Detection via Temporal Graph Learning," in Proc. USENIX Security, 2025.

[16] M. Abrar et al., "Reproducibility in Provenance-based Intrusion Detection Systems," ACM Comput. Surv., 2025.

[17] MITRE, "MITRE ATT&CK®," 2024. [Online]. Available: https://attack.mitre.org/

[18] M. Abdin et al., "Phi-3 Technical Report," Microsoft Research, Tech. Rep. MSR-TR-2024-12, 2024.

[19] Meta AI, "Llama 3.2: Revolutionizing edge AI and vision with open, customizable models," 2024. [Online]. Available: https://ai.meta.com/blog/llama-3-2-connect-2024-vision-edge-mobile-devices/

[20] A. Shekhar, "Step-by-Step Guide to Using Ollama: Local LLM Inference Made Easy," Medium, 2025.

[21] J. Gu et al., "A Survey on LLM-as-a-Judge," arXiv:2411.15594, 2024.

[22] Y. Liu et al., "G-Eval: NLG Evaluation Using GPT-4 with Better Human Alignment," in Proc. EMNLP, 2023, pp. 2511–2522.

[23] L. Zheng et al., "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena," in Proc. NeurIPS, 2023.

[24] D. Preuveneers et al., "The Paradigm of Hallucinations in AI-driven Cybersecurity Systems," Comput. Electr. Eng., vol. 124, 2025.

[25] A. Alansari and H. Luqman, "Large Language Models Hallucination: A Comprehensive Survey," arXiv:2510.06265, 2025.

[26] OWASP Gen AI Security Project, "LLM01:2025 Prompt Injection," 2025. [Online]. Available: https://genai.owasp.org/llmrisk/llm01-prompt-injection/

[27] Y. Liu et al., "Prompt Injection Attack against LLM-Integrated Applications," arXiv:2306.05499, 2023.

[28] K. Zhang et al., "Joint Face Detection and Alignment Using Multi-task Cascaded Convolutional Networks," IEEE Signal Process. Lett., vol. 23, no. 10, pp. 1499–1503, 2016.

[29] J. Guo and Z. Zhang, "Sample and Computation Redistribution for Efficient Face Detection," arXiv:2105.04714, 2021.

[30] J. Deng et al., "ArcFace: Additive Angular Margin Loss for Deep Face Recognition," in Proc. IEEE/CVF CVPR, 2019, pp. 4690–4699.

[31] J. Deng et al., "InsightFace: An Open Source 2D&3D Deep Face Analysis Library," arXiv:2107.04538, 2021.

[32] M. T. H. Fuad et al., "Recent Advances in Deep Learning Techniques for Face Recognition," IEEE Access, vol. 9, pp. 99112–99142, 2021.

[33] K. He et al., "Deep Residual Learning for Image Recognition," in Proc. IEEE CVPR, 2016.

[34] N. E. Khalifa, M. Loey, and S. Mirjalili, "A Comprehensive Survey of Recent Trends in Deep Learning for Digital Images Augmentation," Artif. Intell. Rev., vol. 55, pp. 2351–2377, 2022.

[35] Z. Yu et al., "Deep Learning for Face Anti-Spoofing: A Survey," IEEE Trans. Pattern Anal. Mach. Intell., vol. 45, no. 5, pp. 5609–5631, 2023.

[36] W. J. Scheirer et al., "Toward Open Set Recognition," IEEE Trans. Pattern Anal. Mach. Intell., vol. 35, no. 7, pp. 1757–1772, 2013.